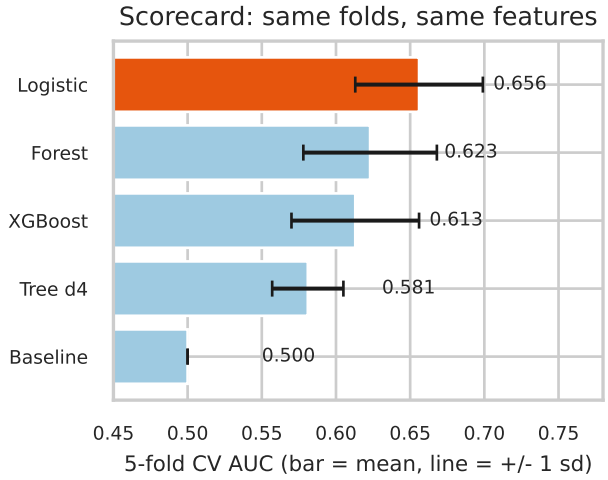
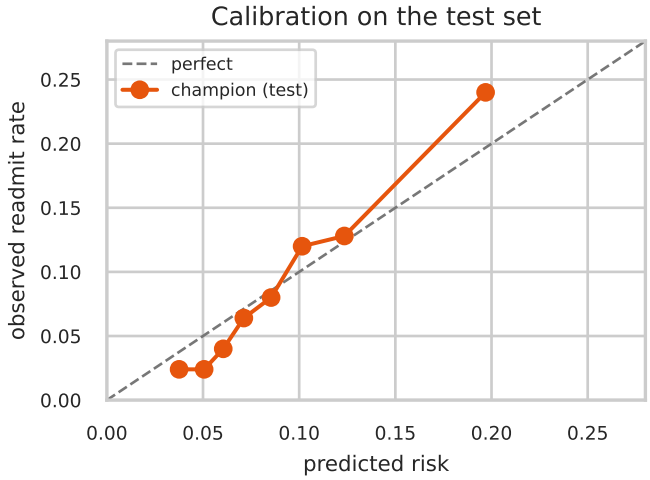
Day 21 Cheat Sheet: The Week 3 Model Scorecard

Week 3 wrap-up: compare every model fairly, pick by a written rule, test once, save it

TERM	WHAT IT MEANS	CODE
Fair comparison	same features, same split, same CV folds for every model	StratifiedKFold(5, ...)
Baseline	predicts the base rate; the floor every model must beat	DummyClassifier("prior")
cross_validate	several metrics per fold in one call	cross_validate(m, X, y, ...)
ROC AUC	ranking quality: does a readmit score above a non-readmit	scoring="roc_auc"
PR-AUC	precision-recall summary; base rate = 0.09 here	"average_precision"
Brier score	mean squared error of probabilities; lower is better	"neg_brier_score"
Fold sd	how much a score wobbles between folds; your noise ruler	r["test_auc"].std()
Selection rule	written before testing; prefer simpler if within 1 sd	if gap < sd: simpler
Hold-out test	opened once, for the champion only	predict_proba(X_test)
Model card	what, data, metrics, intended use, limits	joblib.dump + json





THIS DATA: the same 4,000 simulated discharges as days 14-20 (seed 14), 6 features, readmit rate 9%
5-fold CV AUC: logistic 0.656, random forest 0.623, XGBoost 0.613, tree 0.581, baseline 0.500
Fold sd is about 0.04, so logistic vs forest (gap 0.033) is a lean, not a landslide
Champion: Logistic regression. Test once: AUC 0.723, PR-AUC 0.231, Brier 0.077
Top 20% flagged for a follow-up call: catches 41% of readmissions at 18% precision (base rate 9%)

HABITS THAT KEEP YOU HONEST
- Always include a baseline row; a model that cannot beat it is not a model
- Compare on the same folds; report mean AND sd, not just the winner's number
- Fancier is not automatically better: the true signal here is additive, so logistic fits it well
- Write the selection rule first, then open the test set exactly once
- Test AUC above CV AUC (0.72 vs 0.66) is luck of one 1,000-row split; quote CV as your estimate

```
cv = StratifiedKFold(5, shuffle=True, random_state=21)
r = cross_validate(model, X_train, y_train, cv=cv,
        scoring={"auc": "roc_auc", "pr_auc": "average_precision"})
joblib.dump(final, "readmit_champion.joblib")
```

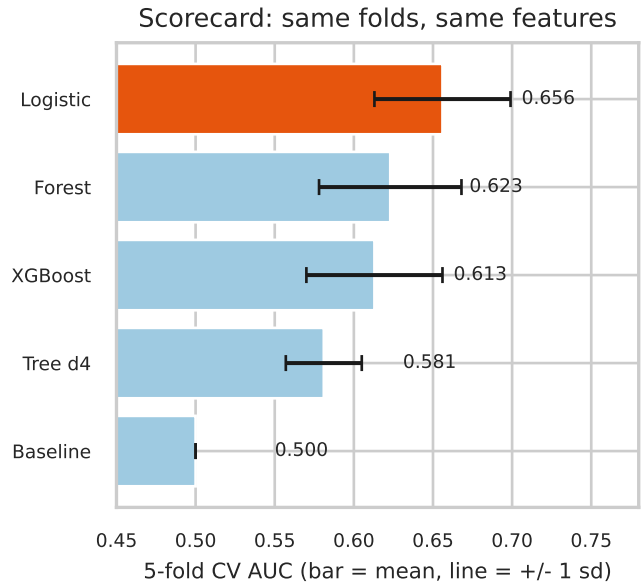
Infographic: Five Models, One Fair Race

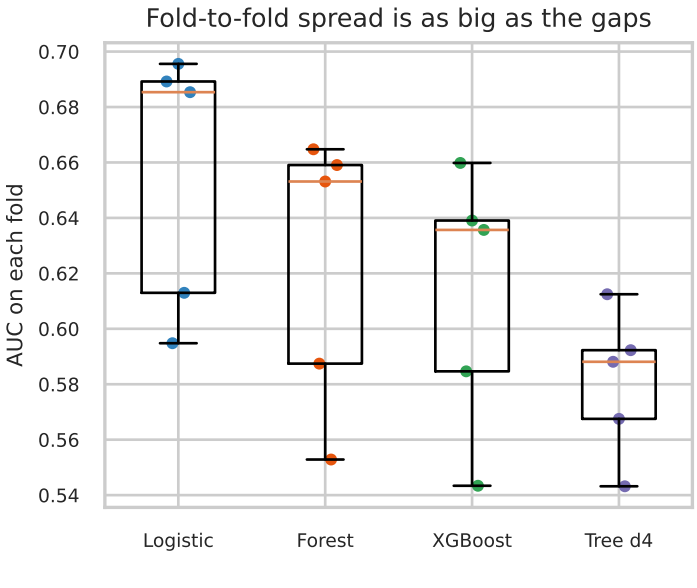
Week 3 recap on the same simulated readmissions data

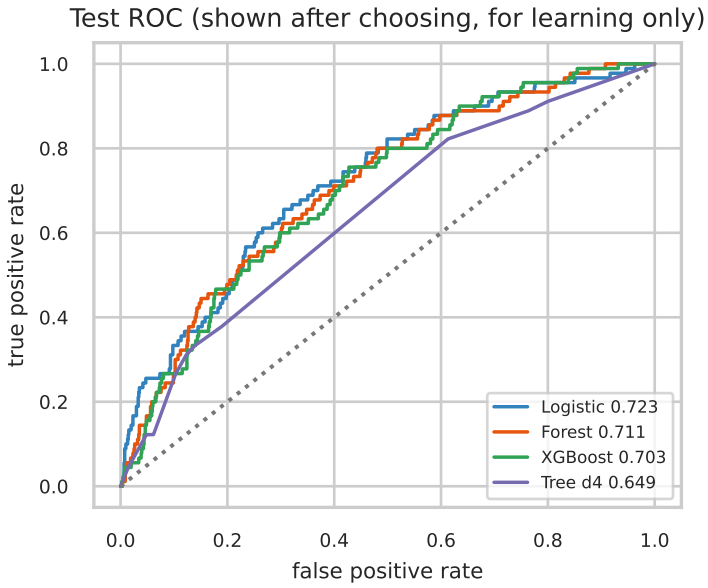
CHAMPION
Logistic

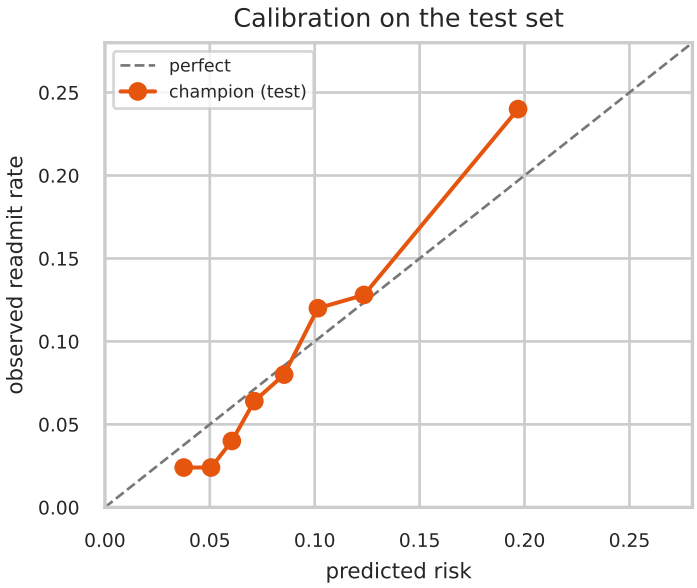
CV AUC
0.656

TOP 20% CATCHES
41%









Takeaways
- Week 3 tour: trees (day 15), forests (16), XGBoost (17), thresholds (18), clusters (19), PCA (20)
- One scorecard with a baseline, mean and sd beats a pile of separate notebooks
- The simplest model won because the true risk is a smooth add-up of a few factors; boosting shines on messier data
- Pick by a rule, test once, then save the model with a model card saying what it is for and what it is not
- Tomorrow: week 4 opens with regression, predicting a number (length of stay) instead of a yes or no

Hands-On Worksheet: ds-genai-90-days / day21

60-90 minutes | notebook: day21/model_scorecard.ipynb + day21/README.md

[ ] mkdir day21, conda activate ds90, open day21/model_scorecard.ipynb

[ ] Rebuild the days 14-20 data and the same split (snippet 1)

[ ] Define the 5-model line-up, baseline included (snippet 2)

[ ] Run cross_validate with AUC, PR-AUC, Brier on the same folds (snippet 3)

[ ] Plot the scorecard bar chart with +/- 1 sd error bars

[ ] Box-plot the per-fold AUCs; is the top gap bigger than the sd?

[ ] Write your selection rule in a markdown cell BEFORE step 8

[ ] Pick the champion and score the test set once (snippet 4)

[ ] Draw the calibration curve; is the model over- or under-confident?

[ ] Top 20% flag: what share of readmissions do you catch?

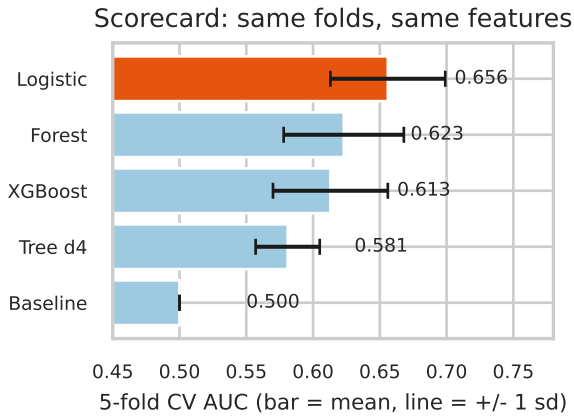
[ ] Save the model + model_card.json, reload, check (snippet 5)

[ ] README: scorecard table, 3 charts, champion and why, limits

[ ] git add day21 && git commit && git push (snippet 6)

[ ] Stretch: log the scorecard to a Snowflake table (snippet 7)

```
r = cross_validate(model, X_train,
      y_train, cv=cv, scoring=scoring)
r["test_auc"].mean(), r["test_auc"].std()
```



CV AUC logistic: _____ forest: _____ XGBoost: _____ tree: _____ baseline: _____

Champion: _____ Test AUC: _____ Brier: _____ Top 20% catches: _____ %

Day 21 Self-Check Quiz

1. Why must every model use the same CV folds in a scorecard?

_____

2. What does the baseline (DummyClassifier) row tell you, and what AUC does it get?

_____

3. Logistic beats the forest by 0.03 AUC with fold sd about 0.04. How would you describe that gap?

_____

4. Why might logistic regression beat XGBoost on this data?

_____

5. Why write the selection rule before you open the test set?

_____

6. Test AUC is 0.72 but CV AUC was 0.66. Which do you report as the estimate, and why?

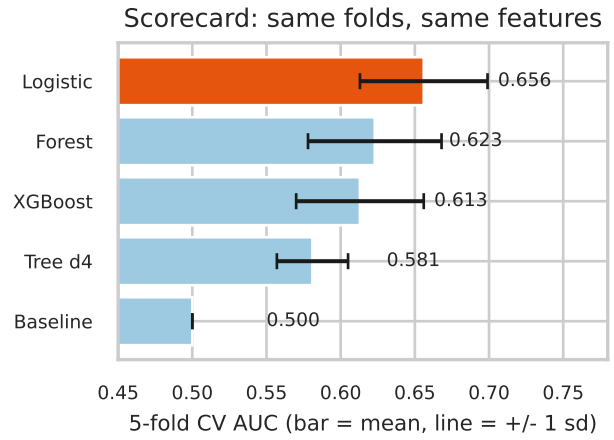
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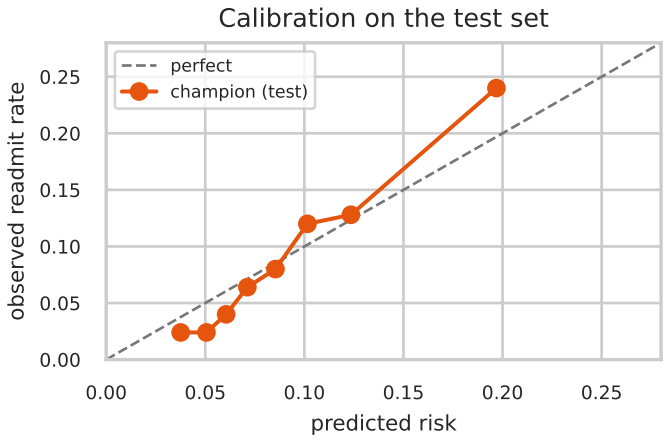
7. Brier score vs AUC: what does each one measure?

_____

8. Name three things a model card should include.

_____





Answers:
1 so differences come from the models, not from which patients landed in which fold
2 the floor: it predicts the 9% rate for everyone, so it cannot rank anyone; AUC 0.500
3 a lean, not a clear win: the gap is smaller than one fold sd, so pick the simpler model and say so
4 the true risk here is an additive mix of a few factors, exactly what logistic models; boosting adds variance without new signal
5 so the test set stays an honest, untouched check; choosing after peeking turns it into another training set
6 CV (0.66): it averages 5 folds; one 1,000-row test split can be lucky or unlucky
7 AUC measures ranking (who is riskier); Brier measures how accurate the probabilities themselves are
8 any three of: model and data, target and features, metrics, intended use, out-of-scope use, limits, version